



SWE 450 : Thesis Report

WoNBias 2.0: A Refined & Expanded Dataset for
Classifying Bias Prejudice Against Women in Bengali Text

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The thesis entitled "WoNBias 2.0: A Refined & Expanded Dataset for Classifying Bias & Prejudice Against Women in Bengali Text" submitted by **Naimur Rahman (2019831079)** and **Kh Mahmudul Hassan (2018831039)** on 28th November 2024 is, hereby, accepted as the partial fulfilment of the requirements for the award of their Bachelor of Engineering Degrees.

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Certificate of Acceptance

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ABSTRACT

In this digital era, people are continuously engaging with the digital world, especially when expressing their opinions on different online platforms. People regularly share their good or bad opinions on these social sites. Large Language Models like OpenAI’s ChatGPT, Meta’s LLaMA, Gemini, and Claude have been making continuous efforts to address and mitigate all kinds of biases. Significant progress has been made in reducing the biases that people face on online platforms, especially on social media. Major tech giants like Google, Meta, Snapchat, and Microsoft have contributed to this field of bias reduction. Most of these contributions, however, are focused on the English-speaking world, aiming to create secure, peaceful, and safe online platforms for everyone. Unfortunately, Bengali-speaking netizens are not safe in the online world. Aggressiveness and bias in many sectors are still common occurrences. In particular, hateful comments and biased attitudes toward women remain prevalent in our society. Bangladeshi women are not safe from hate speech.

In our research, we are focusing on classifying all kinds of bias, hateful comments, demotivating text, and stereotypes faced on online platforms. Bengali large language models (LLMs) still have significant progress to make compared to the LLMs developed by major tech companies in English. We focused on mitigating and detecting this kind of hateful speech and bias against women. The original dataset "WoNBias: a dataset to classify bias & prejudice against women in Bengali text", contains 11,178 entries covering bias and prejudice of women in Bengali text. We expanded the dataset through the addition of 10,010 new entries. With this addition, the dataset has grown to a total 21,188 entries. We collected part of our dataset by surveying the general population, where more than 90% of the respondents were women. The rest of our data came from a variety of sources like Facebook group posts, comments, news articles, blog posts, etc. We have collected 10,010 Bangla sentences and short paragraphs, dividing them into three categories: positive (respectful to women), negative (disrespectful to women), and neutral (not related to this). Most of the data was collected manually. By using this dataset, we hope to create a platform to better detect biases and prejudices that can be used for a range of purposes, such as training less biased language models, advanced content filtering, and hateful comment flagging.

Keywords: Large Language Model, Gender Bias, Bias, Data Collection, Classification

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1. INTRODUCTION

1.1. Motivation

Gender discrimination, or sexism, manifests in various ways across society. Language plays a significant role in perpetuating gender discrimination by reinforcing stereotypes and societal hierarchies. In everyday communication, women often experience bias and social exclusion, facing both subtle prejudices and overt mental abuse within existing social systems. This discrimination creates a challenging environment for women. Globally, women encounter unequal treatment in many areas, including the workplace, educational institutions, public transport, social programs, national issues, marketplaces, online platforms, and even within their own families.

To detect and address "Linguistic Harassment" or any form of harassment, social experts, organizations, responsible authorities, social workers, human rights advocates, and civil society are taking various proactive measures. Large Language Models (LLMs) can be particularly useful in detecting gender biases. These models have gained global recognition for their advanced natural language understanding and generation capabilities. LLMs exhibit sophisticated language comprehension, the ability to generate human-like text, strong contextual awareness, and effective problem-solving skills, making them highly valuable across a range of applications.

Several prominent LLMs, such as OpenAI's ChatGPT, Meta AI's LLaMA, Databricks' Dolly 2.0, and Bangla Bert were developed and released, quickly gaining widespread attention. Numerous studies have explored the detection of sexual harassment in conversations and on social media. However, the challenge remains whether LLMs can consistently identify and address such complex issues, particularly in sensitive domains like sexual harassment. Given the models' sensitivity to input variations, it is essential to evaluate whether LLMs can accurately capture the nuanced linguistic subtleties present in these specific contexts.

The primary aim of this paper was to expand the dataset to include a wide range of stereotypical, hateful, or discouraging comments and ideologies directed at women. The original dataset, WoNBias: A Dataset for Classifying Bias & Prejudice Against Women in Bengali Text, consists of 11,178 entries focused on identifying bias and prejudice against women in Bengali text. In this work, we extended the dataset by adding 10,010 new entries, resulting in a total of 21,188 entries. In this study, we focused on and analyzed various online sources, including Facebook posts related to events involving women, Facebook groups promoting hostility toward progressive women, YouTube content related to women, blogs, Quora discussions, Instagram, TikTok, news articles covering incidents of discrimination and violations against women, as well as blog posts where individuals

shared their ideologies on these issues.

During the data collection process, we followed a structured set of guidelines to ensure that sentences or paragraphs were classified as negative or positive only if they clearly fit into those categories. Since the majority of the sentences identified were neutral, we decided to include data in the other categories when the meaning was unambiguous and devoid of vagueness.

Throughout the data collection process, we observed that people often incorporate their local culture into their communication, using culturally specific words and expressions. Since culture varies from one society to another, sentiments are often generalized in communication. We also noted that individuals rarely use direct language to harass women; instead, they may employ satirical phrases or sarcasm. Additionally, it was observed that in many cases, people indirectly express their views, making it essential to understand the context to fully grasp the intended meaning. In many cases we observed When events disappear these comments have no meaning in many cases.

1.2. Background Study

Large Language Models (LLMs) have become essential tools in natural language processing (NLP) tasks, including efforts to predict and mitigate bias. However, concerns remain regarding the potential biases embedded within these models. These biases can reinforce stereotypes, contribute to algorithmic unfairness, and undermine trust in AI technologies. LLMs offer an in-depth examination of the societal, ethical, and practical impacts of these biases, emphasizing their influence on data-driven decision-making. Commonly used LLMs include the GPT series (Generative Pre-trained Transformer) [1], BERT (Bidirectional Encoder Representations from Transformers) [2], Meta AI’s LLaMA [3], Databricks’ Dolly 2.0 [4], and BanglaBERT [5]. These models are often employed in examining various types of biases, such as algorithmic bias, data bias, and social bias. For instance, gender bias, a specific form of social bias in NLP, has been widely studied, particularly in the English language [6]. Notable works have investigated gender bias and women’s harassment in NLP applications [7] [8] [9]. Research into gender bias has also extended beyond English, focusing on other languages to address the global nature of the issue [10] [11]. Similarly, LLMs have been applied to explore racial biases in AI interactions, highlighting concerns about fairness and inclusivity [12].

In this study, Bhattacharjee et al. [5] introduce BanglaBERT, a BERT-based model specifically designed for Natural Language Understanding (NLU) tasks. BanglaBERT is trained on 27.5 GB of Bangla text data, which was collected from 110 popular Bangla websites. Using this extensive dataset, the authors developed the BanglaBERT model, aiming to enhance NLU performance for the Bangla language. BanglaBERT dataset focused on Natural Language Inference (NLI) and Question Answering (QA) tasks. The Authors established the BLUB benchmark, a set of evaluation standards or metrics for

assessing the performance of NLU models in Bangla. BanglaBERT, they achieved state-of-the-art results.

In this work, Islam et al. [13] present an annotated sentiment analysis dataset, consisting of informally written Bangla texts. The dataset includes public comments on news and videos from social media, covering 13 domains such as politics, education, and agriculture, labelled as positive, negative, or neutral. There are 15000 social media comments on news and videos and has a mix of dialects and grammatical errors. The authors experiment with various techniques, including linguistic features, recurrent neural networks, and pre-trained models and find that traditional word n-grams outperform other methods, resulting in better classification performance than neural network models in their experiments.

In this research, Kotek et al. [7] explores how large language models (LLMs) respond to gender stereotypes. The findings often provide flawed explanations about sentence structure and grammar to justify their choices. Additionally, LLMs sometimes explicitly reference stereotypes, reflecting biases encoded from the collective intelligence of Western society. The study emphasizes the importance of identifying, understanding, and addressing these patterns to improve LLM behaviour.

In this study, Kurita et al. [14] introduced a template based approach to measure bias in BERT, demonstrating its consistency over traditional cosine-based methods and applying it to evaluate gender bias in Gender Pronoun Resolution. The authors experiment with multiclass settings, such as racial and religious biases.

In this research, Ahn et al. [15] studied ethnic bias in monolingual BERT for English, German, Spanish, Korean, Turkish, and Chinese and developed a Categorical Bias(CB) score on ethnic bias. The authors also proposed a multilingual model and aligned two monolingual models. The authors explored that BERT models reflect the historical and social context of any country.

Abid et al. [16] showed that GPT-3 exhibits a significant and consistent bias linking Muslims with violence. This bias is evident across various tasks like prompt completion, analogies, and story generation. The authors also researched ways to reduce this bias by providing positive words or phrases in the context of the model’s input.

Naous et al. [17] showed multilingual and Arabic monolingual language models display biases favouring entities linked to Western culture. The authors presented a brand-new tool called CAMEL (Cultural Appropriateness Measure Set for LMs), which has 20,368 elements in eight categories that compare Arab and Western civilizations, to-

gether with 628 naturally occurring prompts. In their investigation of 16 distinct language models' cross-cultural performance in Arabic on tasks like sentiment analysis, NER, and story production, the authors found alarming examples of cultural prejudice and stereotyping. The authors revealed that six Arabic pre-training corpora indicate that frequently utilized sources, such as Wikipedia, may not be optimally suited for constructing culturally aware language models if employed in their unmodified form.

Overall, these studies show that various types of bias exist in different forms in Large Language Models (LLM). It is rare to find a training dataset that is completely free from bias. Many training datasets lack sufficient resources for certain languages. However, further study is required to comprehend the mechanisms underlying the connections and to find more effective ways to create and train Large Language Models (LLMs) while minimizing biases of all kinds. Although there are many datasets for English, there are still not enough resources for other languages. Furthermore, bias affects datasets in all languages and is present across a variety of research topics.

By building all language datasets and building models using Large Language Models (LLM) of all languages on bias, we can contribute a lot to the most largely extending research. There is also a huge scope of research on the Bangla language. We can explore the Large Language Model (LLM) on Gender bias as well as women's harassment detection.

1.3. Objectives

1.3.1. Data Collection

The primary goal of this research is to collect a large amount of data. The dataset will consist of hate speech, offensive language, abusive words, derogatory speech, bias and prejudices toward women. The amount of positive and neutral data will be same as negative data.

1.3.2. Ensure Diversity & Authenticity

The next step for this study will be to ensure that this data comes from varied and definitely trustworthy sources. It will include online surveys, group discussions, targeted online communities, personal blogs and local newspapers. These will help us maintain a diverse and authentic dataset at all stages of the pipeline.

1.3.3. Categorize Data

After collecting the data, they will be carefully categorized into three predefined categories: Neutral (0), Positive (1), and Negative (0). The manual categorization guarantees structural data to get the basis for training the model. It will also improve the overall

readability of the dataset. These steps are important in ensuring the accuracy of the data and getting the most out of the model when training it.

1.3.4. Data Quality Analysis

Once the dataset is created, categorized, and shuffled, then it is to be divided into several chunks among researchers and assistants to go through again and check for any mislabeled sentences or paragraphs. Upon finishing that the dataset will be combined in one place again and again it will go through another check by the researchers. Any other languages, misspelling, use of emojis, and conflicting or confusing data will be dropped through this process.

1.3.5. Training & Evaluation

The next objective is to divide the dataset into different proportions, train distinct models on these subsets, and interpret the results in order to determine the best configuration for the models. After this stage, the research seeks to present particular measurements designed to assess and benchmark the models against one another, so that the best combination for the task is found

2. DATA COLLECTION

2.1. Introduction

When we started collecting data, we noticed that we were gathering only a very small amount of related data from a single source. So, we decided to collect data from every possible source, both offline and online. This approach helped us understand the sources better. As a result, we were able to collect our dataset from a diverse range of origins.

2.2. Sources & Participants

When we were collecting data from multiple sources, we tried to engage women from various backgrounds and experiences. Also, we targeted to collect data from online communities of both pro-feminine and anti-feminine groups.

2.2.1. Participants

Firstly, we released the form in various universities, including the University of Rajshahi, American International University of Bangladesh, Begum Rokeya University, Bangladesh Agricultural University, Islamic University, Pabna University of Science and Technology, and Patuakhali University of Science and Technology. In the form, we asked questions about their negative experiences as girls or women. Then, we shared the form with people we were familiar with. Lastly, we distributed the form on our campus, Shahjalal University of Science and Technology.

The individuals who filled out the form were primarily college students, university students, corporate professionals, homemakers, and both married and unmarried girls and women. A significant portion of the negative dataset was gathered from their responses.

2.2.2. Other Sources

We collected data from multiple sources, with the majority of our dataset coming from online communities. We began collecting positive and negative data simultaneously. A significant portion of the negative dataset was gathered from comments on YouTube videos. We specifically selected videos that portrayed negative attitudes or biases against women, ensuring that the comments aligned with this tone. This targeted approach allowed us to capture relevant data that highlighted societal prejudices against women. Then, we started collecting negative sentences from Facebook pages, communities, and groups. To gather negative sentences, we focused on anti-feminine Facebook groups. These groups are predominantly composed of male members.

[Form Link](#)

FORM QUESTIONS

- শিক্ষা প্রতিষ্ঠানে শোনা একটি বাক্যের উদাহরণ দিন , যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - কম্পিউটারের বিষয়গুলো মেয়েরা একটু কমই বুঝে।
- কোনো সামাজিক অনুষ্ঠানে (বিয়ে, ঘুরতে যাওয়া) শোনা কথা, যা আপনাকে(নারী) বা কোনো নারীকে উদ্দেশ্য করে বলা এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - এত চাকরী করে কি হবে! মেয়ে হইসো, সংসারের খেয়াল রাখা শেখো আগে।
- পারিবারিক সদস্য, প্রতিবেশী বা আত্মীয়স্বজন থেকে আসা কোনো কথা কিংবা বৈষম্যমূলক আচরণ , যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - তুমি মেয়ে, তুমি কম খেলেও হবে।
- পরিবার থেকে আকাঙ্ক্ষা রাখে এমন কোন কথা, যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - মেয়েদের পড়িয়ে লাভ নাই, দুদিন পর তো শশুর বাড়ি চলে যাবে।
- কর্মক্ষেত্রে শোনা একটি বাক্যের উদাহরণ দিন , যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - মেয়েদেরকে শেখাতে বেশি সময় লাগে।
- বৈবাহিক সম্পর্কের জন্য তৈরী হওয়া পরিবেশে (শশুর বাড়ি) একটি বাক্যের উদাহরণ দিন , যা আপনাকে(নারী) বা কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য করেছেন?
উদাহরণ - যতই চাকরি কর, রান্নার দায়িত্বটা কিন্তু ঘরের মেয়েরই।
মেয়েদের দায়িত্ব হলো ঘর সামলানো।

Figure 2.1: Form Questions Image

A significant portion of our negative dataset was sourced from the Facebook pages of Bangladeshi female celebrities.

We collected positive datasets from a variety of sources. A significant portion of this data was gathered from YouTube videos highlighting women's achievements and empowerment. Additionally, we utilized articles and blogs from various newspapers that focused on topics such as women's rights, women's empowerment, and their contributions to society. We also sourced data from events and publications related to International Women's Day, which celebrated women's successes across different fields. News articles

FORM QUESTIONS

- বিভিন্ন সামাজিক যোগাযোগ মাধ্যমে (ফেসবুক , ইউটিউব) আপনার দেখা বা শোনা মন্তব্য বা স্ট্যাটাস , যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - মেয়েদের দিয়ে সব কাজ হবে না ।
- পোশাকের কারণে এমন কোন কথা শুনেছেন বা অভিজ্ঞতা অর্জন করেছেন, যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - বোরকা পরে বিশ্ববিদ্যালয়ে আসার কী দরকার, ঘরে থাকলেই হয় । যে পোশাক পরেছেন , না পরলেও হত।
- বিভিন্ন যানবাহনে এমন কোন কথা শুনেছেন বা অভিজ্ঞতা অর্জন করেছেন, যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - সব জায়গায় নারী পুরুষ সমান অধিকার, তাহলে বাসে সিট না পেলে দাঁড়িয়ে থাকতে সমস্যা কী ?
- উচ্চশিক্ষার ক্ষেত্রে বিরূপ মনোভাব তৈরী করে এমন কোন কথা শুনেছেন বা অভিজ্ঞতা অর্জন করেছেন, যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - মেয়েদের বিশ্ববিদ্যালয়ে পড়ার কি প্রয়োজন, এরা দিনশেষে ঘরই সামলাবে
- মেয়েদের খেলাধুলার বিষয়ে আপনি এমন কোন কথা শুনেছেন বা অভিজ্ঞতা অর্জন করেছেন, যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?
উদাহরণ - তুমি মেয়েমানুষ বাহিরে খেলাধুলা করছো কেন ?
- উপরে উল্লেখিত অভিজ্ঞতা ছাড়াও ভিন্ন কোন অভিজ্ঞতা যা আপনাকে(নারী) বা পরিবারের কোনো নারীকে উদ্দেশ্য করে বলা হয়েছে এবং তাতে লিঙ্গপ্রসূত পক্ষপাত অথবা লিঙ্গের ভিত্তিতে বৈষম্য লক্ষ্য করেছেন?

Figure 2.2: Form Questions Image

played a crucial role in our positive data collection. A considerable amount of data was collected from personal blog sites where individuals shared their experiences and opinions on various topics related to women's rights, empowerment, and societal roles.

Finally, we began collecting neutral data. When we were collecting positive and negative sentences, we found some sentences that were neither positive nor negative. We count them as neutral data. The dataset provided by csebuetnlp/BanglaParaphrase [18] on Hugging Face was manually filtered to get the remaining neutral data.

2.3. Data Collection Statistics

After thoroughly exploring numerous online groups, analyzing comments on YouTube videos, and reviewing online posts from various platforms, including social media, blogs, newspaper articles, and existing datasets on Bengali paraphrases [18], we successfully completed our data collection process. This approach helped us to gather a diverse range of dataset. The previous dataset, "WoNBias: A Dataset for Classifying Bias & Prejudice Against Women in Bengali Text", consists of 11,178 entries. We obtained permission from the owner of the dataset to use it in our research [1]. Thus final size of our dataset stands at 21,188. The data overview is presented in Table 2.1, Table 2.2 showing the number of data on all three of the categories.

Category	Number of Sentences
Negative	2883
Positive	3809
Neutral	3318
Total	10,010

Table 2.1: Data Overview of Newly Collected Dataset

Category	Number of Sentences
Negative	6911
Positive	7315
Neutral	6962
Newly Collected	10,010
Total	21,188

Table 2.2: Data Overview of Complete Dataset

2.4. Data Quality

Our entire data collection process was manual. As we collected the dataset manually, the possibility of any ambiguous data is almost non-existent.

2.4.1. Data Collection Guidelines

To measure the quality of the collected data, we established a set of rules about the characteristics that should be followed. These rules included:

[Drive link of the permission.](#)

1. As long as there is no prejudice towards any certain group, neutral data can be either positive or negative.
2. Variations of the same sentence structure can be accepted but duplication must be avoided.
3. Positive sentences must either explicitly encourage women or highlight their hardships and victimization in a supportive manner.
4. Make sure to capture all types of terms used to refer to women, whether they are positive or negative.
5. Search for any derogatory, hateful, biased, prejudiced, or discouraging words and phrases that are widely recognized and understood across different contexts. These can include abusive language, threats, comparisons to unclean or undesirable objects, and suggestions related to self-harm or suicide. Additionally, look for phrases that demean women by comparing them as inferior to men or other groups, labeling them as failures, or making negative comments about their physical appearance or body parts.
6. To be included in the dataset, sentences and paragraphs must be self-explanatory of their category. They cannot depend on any outside context to determine whether they are prejudiced or derogatory toward any women.

3. METHODOLOGY

3.1. Mindset Behind the Approach

This research aims to develop a robust model to detect and label social and personal biases, prejudices, and hatred that women face regularly in Bengalispeaking communities. During the rise of large language models, other researchers from mostly Bangladesh are planning to develop Bengali language models as well. As we know, these language models depend heavily on huge datasets curated from all kinds of social and news platforms, and there remains the chance of unfiltered derogatory data may also enter the training phase. This can happen because there are not many tools for moderation to cover all kinds of biases and hate speech online targeting other people and their personal lives. We aim to combat a part of this problem by detecting social and personal biases toward women in our society, to begin with. We reviewed over a thousand online posts from different social sites, blog sites, newspaper articles, and existing datasets on Bengali-paraphrase [18] to extract comments, posts, rules, activists’ charter, government plans, and essays to create a complete and diverse dataset that can be labeled as gold standard.

3.2. Data Collection

We collected our dataset from multiple sources, as previously described in the Data Collection Chapter. Our data collection sources were YouTube video comments, Facebook pages, communities, groups, news articles, personal blogs, online posts from different social sites, blog sites, and the existing dataset on csebuetnlp/BanglaParaphrase [18]. We were conscious about anyone’s personal information while collecting data. We made sure to avoid collecting any personal information from any of the data sources. We were careful to follow ethical guidelines and respect privacy throughout the entire data collection process.

3.3. Measuring Data Quality

As mentioned earlier, our goal was to create a highly reliable and well-curated dataset. In the fields of machine learning and natural language processing (NLP), having a trustworthy dataset is crucial for accurately evaluating the performance of algorithms and models. With this in mind, we carefully designed and gathered our data to ensure its quality. Throughout the process, we focused on meeting several important criteria to make sure the dataset would be suitable for training and testing the models effectively.

3.3.1. High-Quality Annotations

We carefully reviewed our dataset multiple times to ensure high-quality annotations. Each annotation was made focusing on achieving the highest possible accuracy and reliability. We tried to ensure the dataset met the necessary standards for research and analysis. Our commitment to quality was central throughout the annotation process.

3.3.2. Filtering Data

After completing the collection, we divided up the entire dataset into multiple smaller parts and started reviewing them. Then we sat together, reviewed again, and dropped any vague data.

3.3.3. Representation and Diversity

We gathered our data from a diverse range of sources to capture various scenarios and variations in sentences. To ensure this diversity, we released our forms in schools, colleges, and universities, as well as among people we were familiar with. This approach allowed us to collect data from multiple sources, ensuring a rich and varied dataset. As a result we were able to collect data from diverse mentalities.

3.3.4. Consistency

We set some guidelines(mentioned in data collection process 2.4.1) for data collection that is required to preserve the consistency in the collected data while we were collecting data. By following the guidelines properly, we ensured the pattern of data was the same even though they were collected separately.

3.4. Identifying model to develop

Our main target was to find such a model that can give more accuracy and predict accurately. To find such a model, we searched for existing natural language processing model for Bengali language. We read several papers and checked any existing work done using these classification model. In our findings, we observed some models, including csebuetnlp/banglabert's [5] sequence classification model, SetFit for Text Classification [19] and SetFit Text Classification Hyper-parameter Search [19]. Finally, we decided to use csebuetnlp/banglabert's [5] sequence classification model to develop our dataset. Each model has their own capabilities and complexities. BanglaBert is specifically built for dealing with Bengali language. We aim to optimize the performance metrics such as accuracy, precision, recall, or F1- score to the best of our ability. Also, there are some classification works already done using this model like "SentNoB: A Dataset for Analysing Sentiment on Noisy BanglaTexts" [13]. So judging by the interpretability of this model we deduced that it would be crucial for understanding the underlying features of our dataset well enough to achieve our target.

3.5. Identifying model for comparison

We trained our model using csebuetnlp/banglabert’s [5] sequence classification model. Then we planned to choose a real-world application. As previously the dataset was tested by OpenAI GPT-3.5 Turbo model, we choosed Gemini 1.5 pro as the second model. We compared the result for the same test dataset.

Gemini 1.5 Pro is known for its advanced capabilities in handling natural language processing tasks, including sequence classification. Gemini 1.5 Pro supports multilingual capabilities, and its adaptability to specific languages makes it a good choice for testing datasets in Bengali. Its architecture is optimized to handle large datasets effectively, ensuring better scalability and consistent performance during inference. It is designed to handle real-world applications, making it a practical choice for our dataset derived from social media and news platforms. By comparing the results of BanglaBERT with Gemini 1.5 Pro, we introduce diversity in the testing process. This helps evaluate how different architectures and pre-training approaches impact sequence classification results.

4. RESULTS AND ANALYSIS

4.1. Training & Testing the models

4.1.1. Enactment and Evaluation of Intended Model

Using our dataset, we enacted the csebuetnlp/banglabert's [5] sequence classification model for double sequence classification to classify sentences and paragraphs with the label ranging from 0 to 2, where 0 stands for neutral, 1 for positive and 2 for negative. The dataset was divided into different ratios for training, validating, and testing. We started with the ratio of -

$$\text{Train : Validate : Test} = 60 : 30 : 10$$

And after adjusting a couple of times, proceeded with the following ratio:

$$\text{Train : Validate : Test} = 80 : 12 : 8$$

Our shuffling ensured a balanced distribution of data across all sets. After dividing, training data stood at 16946, validation at 2544, and testing at 1698. The following figures for train, validate, and test shows the distribution stat -

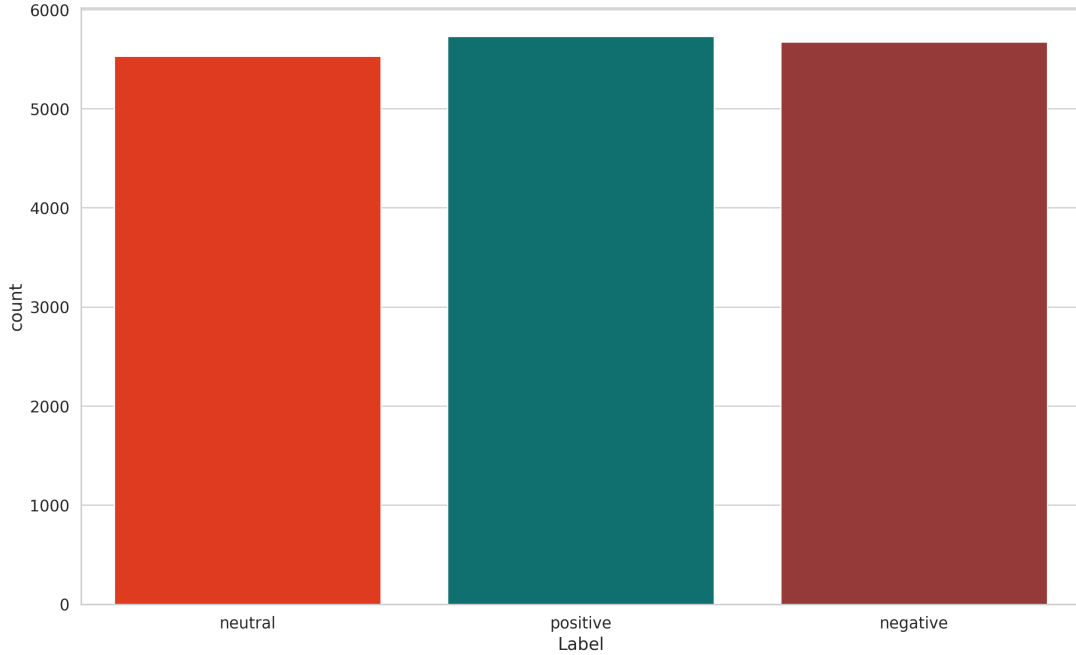


Figure 4.1: Stat for training data

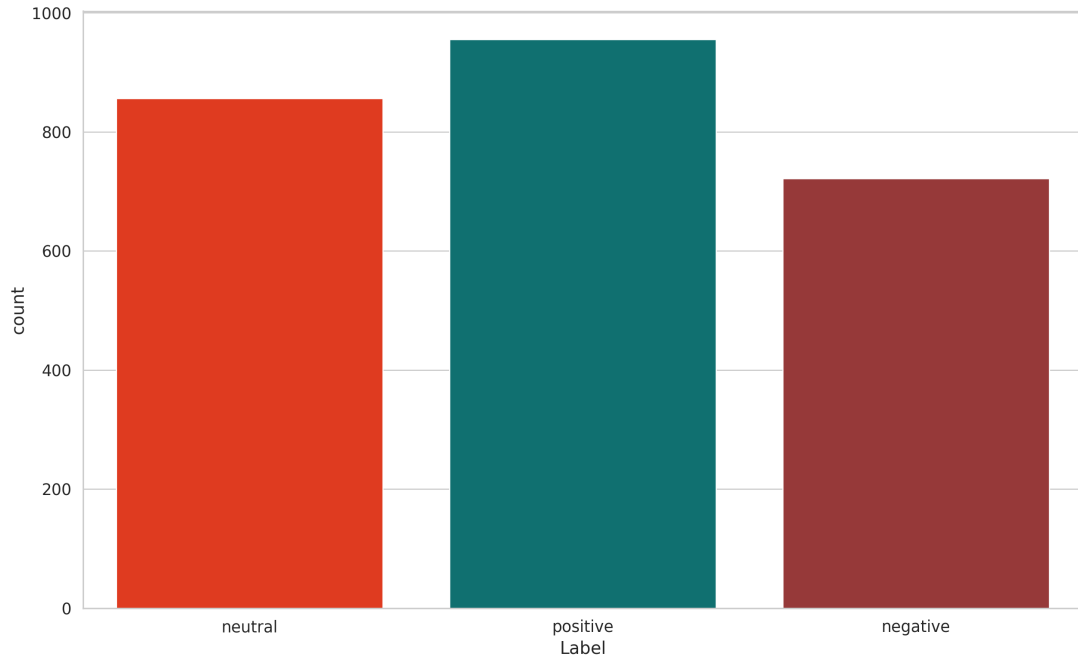


Figure 4.2: Stat for validation data

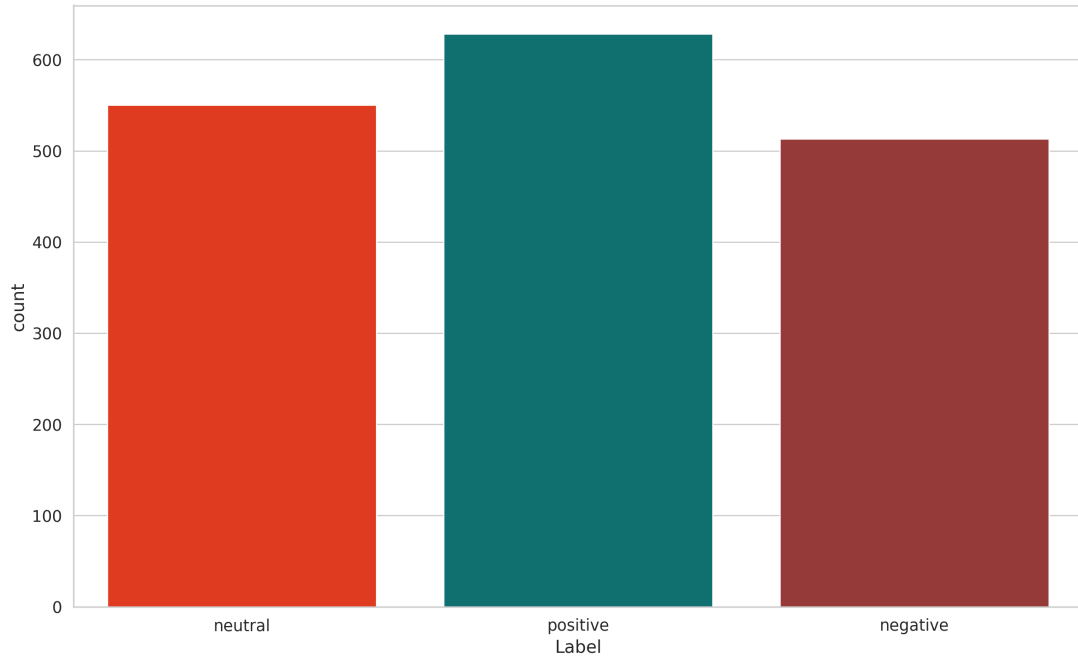


Figure 4.3: Stat for testing data

After this, we used our test dataset of mixed neutral, positive, and negative data on each iteration of our training scenarios. Table 4.1 shows the overall accuracy percentage of the model's prediction on each iteration, indicating a very stable shift on different ratios, ranging from 94.46% to 95.21%. We had to understand that the sentences and paragraphs in our dataset varied very much in length and had constrained length values. So, this

result exceeded our expectations. Still, the statistics showed that the 3rd iteration had the best accuracy overall. An increase in the training dataset size influenced the accuracy better.

Iterations	Category	Ratio	Accuracy
1st iteration	Training	60%	
	Validation	30%	94.46%
	Testing	10%	
2nd iteration	Training	70%	
	Validation	20%	94.84%
	Testing	10%	
3rd iteration	Training	80%	
	Validation	12%	95.21%
	Testing	8%	

Table 4.1: Accuracy on different training scenarios

As the model with the highest accuracy, the 3rd iteration model is kept for later comparison.

4.1.2. Testing the Comparison Model

We sought to test the same testing dataset of biases, stereotypes, and anything hateful or discouraging toward women that was classified by our model, against one of the most sophisticated, cutting-edge advancements in language modeling, characterized by its sophisticated architecture, human-like text generation capabilities, and widespread adoption in the natural language processing community, The Gemini-1.5 Pro model.

We chose this model because it is trained through unsupervised learning from vast amounts of text data and it has acquired a remarkable understanding of language semantics and syntax.

After establishing a connection with its API directly, we designed our prompt to give it an idea of what we are after. We started sending requests with chunks of our test data and started storing the results for comparison in the next stage. During the testing of the Gemini model via its API, 22% of the input sentences were rejected due to safety mechanisms designed to filter potentially harmful content. The remaining 78% of the dataset, which passed the safety checks, was used to evaluate the model’s accuracy. We achieved 69.35% of total accuracy from the Gemini-1.5 Pro model against our test dataset.

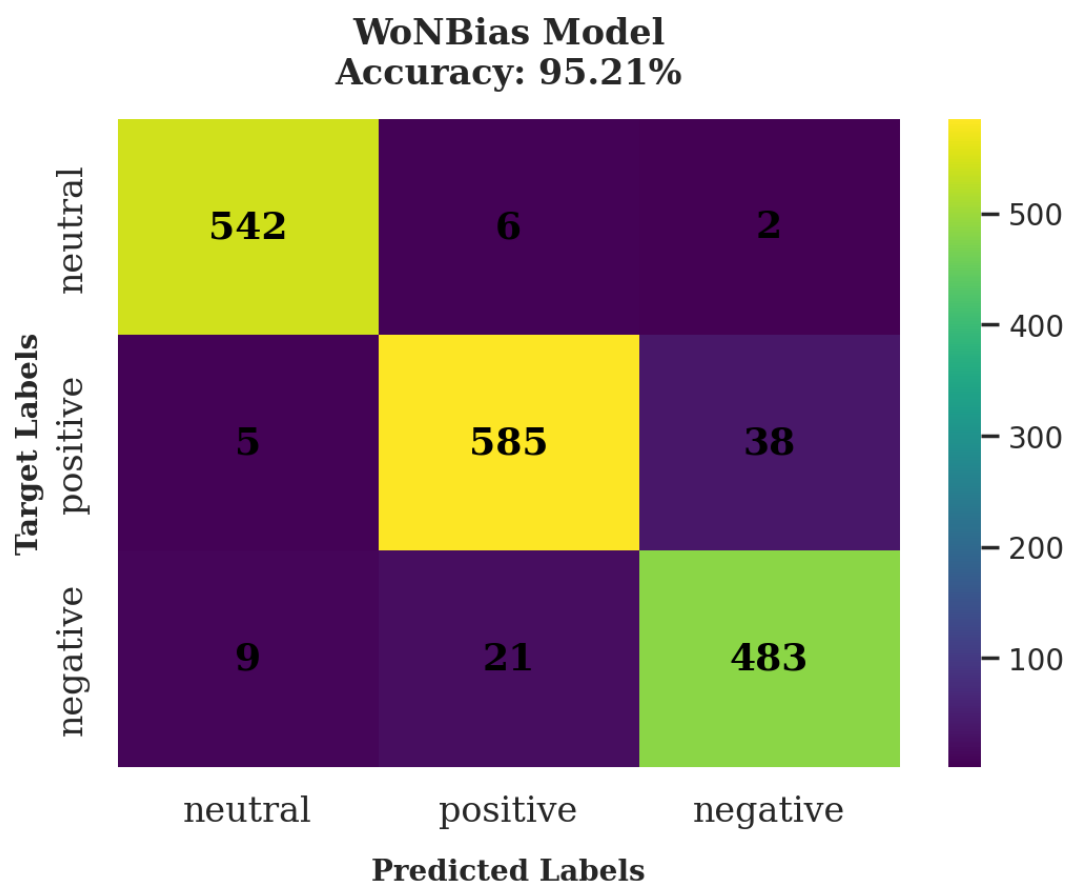


Figure 4.4: Accuracy of the WoNBias 2.0 model

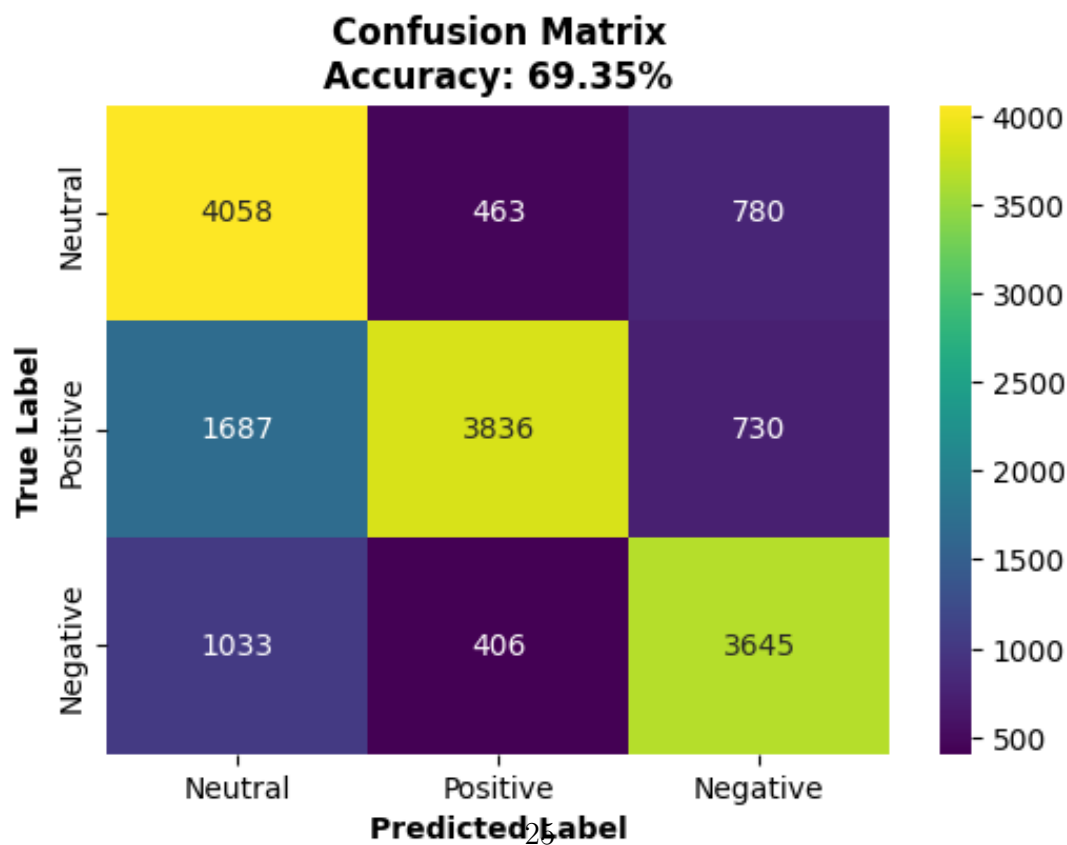


Figure 4.5: Accuracy of the Gemini 1.5 Pro model

4.2. Comparative Analysis and Progress Stat

Judging by the quality of the response received in individual sentences and paragraphs, it was our understanding that the Gemini-1.5 Pro model struggled to classify negative data altogether. When it came to dealing with Bengali syntax and language semantics, this model showed that it has a low understanding of what patterns constitute negative, positive, or neutral data. The difference is clear in the comparison of the two models' Figure 4.8 classifying capabilities. The accuracy drops a significant 26% between the two models.

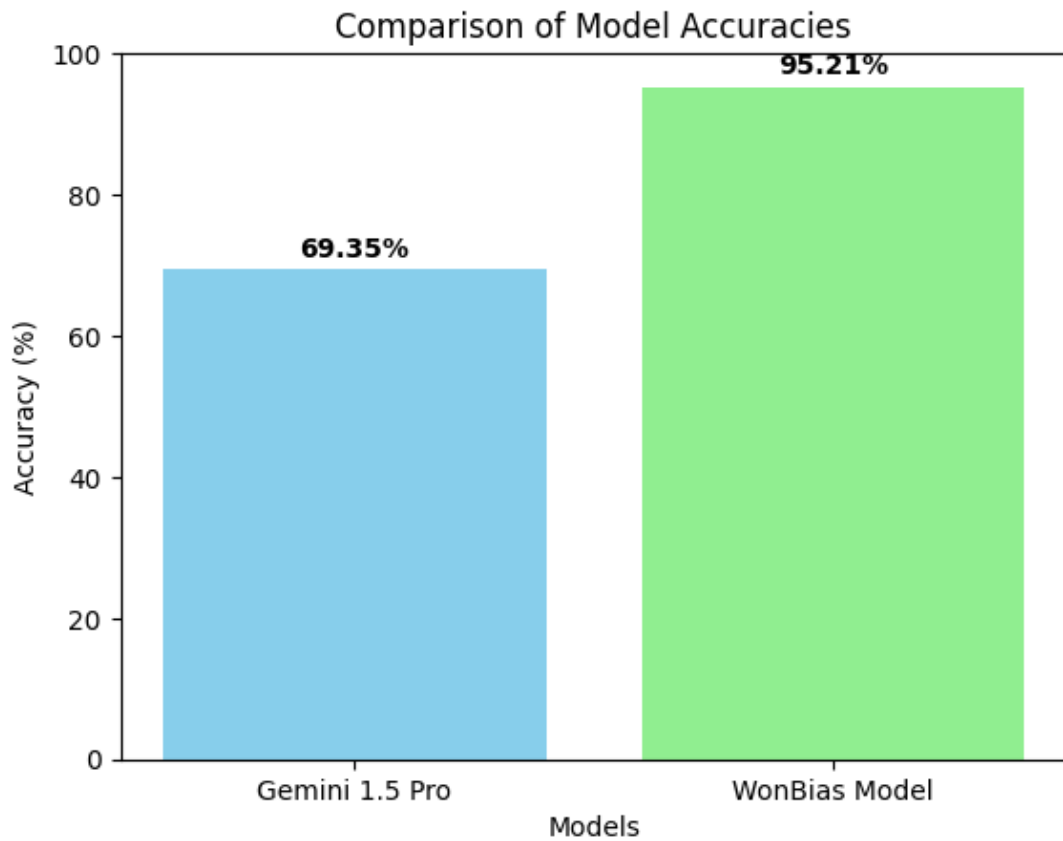


Figure 4.6: Accuracy difference between Gemini 1.5 Pro and WoNBias 2.0 model

Looking at other performance metrics, such as precision, recall, and f1-score, we saw a significant drop in every criterion for the Gemini-1.5 Pro model. This level of difference between the two models indicated a considerable variation in their performance on the classification task.

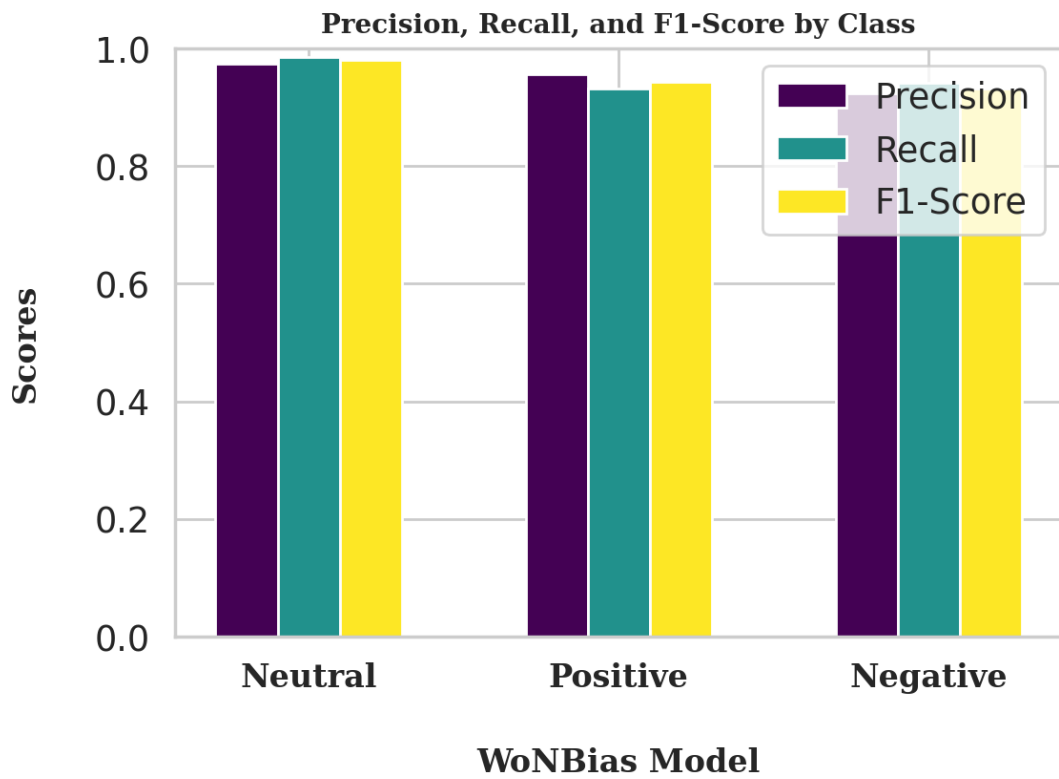


Figure 4.7: Performance stat for WoNBias 2.0 model

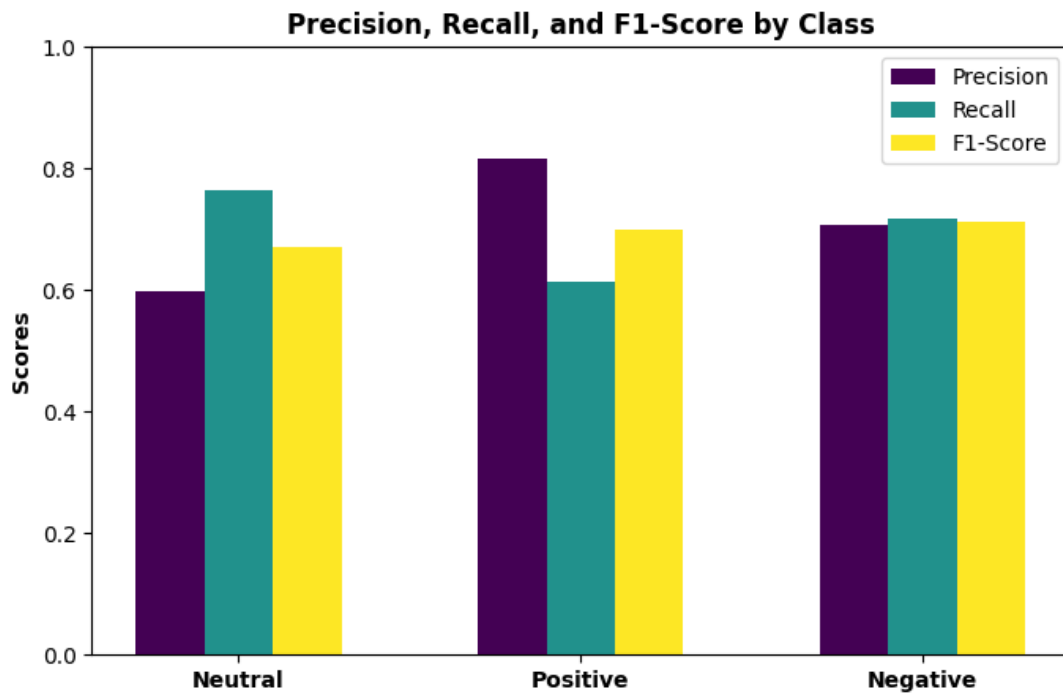


Figure 4.8: Performance stat for Gemini 1.5 Pro model

All evidence suggests that existing trained language models like the Gemini-1.5 Pro lack what it takes to generate moderated content for the Bengali-speaking Demography.

4.3. Limitations

There are several limitations in the process of data collection, training and testing the model that should be taken into consideration.

Firstly, our research was focused solely on hate speech directed against women. However, it is important to acknowledge that individuals of other genders can also be victims of hate speech.

Secondly, we used Gemini 1.5 Pro as our comparison model. However, there are several other large language models (LLMs) that could also serve as comparison models.

Thirdly, as we collected most of our data manually, we could not collect the dataset as per our expectations. We believe that, with additional time, this dataset can grow further.

Lastly, our dataset contains many feminine terms, derogatory. It includes names and pronouns. Our model can't identify feminine names.

5. CONCLUSION

Our trained model proves to be impactful and has a promising effect on the Bengali frontier. In the context of Bangladesh, women typically face hate speech, offensive language, abusive words, derogatory speech, bias, prejudices, discrimination, etc. Our model can identify these types of hateful speech, and it is a crucial first step toward eradicating discrimination against women. This demonstrates the need and importance of a platform that detects this bigotry and its promising impact on the Bengali frontier. Therefore, we tried to achieve that goal by creating a model that can play a vital role in detecting this bigotry. We worked with a vast size of dataset. We dealt with a dataset, size of 21188. It will cover a broader range of biases and prejudices against women if we continuously expand our dataset over the time. This will improve the quality of our model and also provide better moderation tools for those who wish to use it for ethical purposes.

Based on the results of our study, our WoNBias 2.0 model performs well enough to classify a staggering 95% of our test dataset, which covered a balanced number of sentences and paragraphs of all categories. This result even challenges the large language model like the Gemini 1.5 pro model, judging by the differences between accuracy and other performance metrics. Our dataset and model can be used to eliminate biased, hate speech and offensive language. Further research into this matter will only increase the effectiveness of this process.

Although our research has made significant strides, there are still some constraints and limitations. In the future, we will try to test our dataset with other LLMs. Also, we will plan to train and test our model using datasets that include varying amounts of additional noisy data. Additionally, we plan to introduce feminine names to ensure broader coverage. In our data collection process, we will try to engage with a broader range of people and gather data from a variety of diverse and versatile sources.

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